

Summary Data on Acid Cropland and Lime Requirements in Tanzania

GAIA

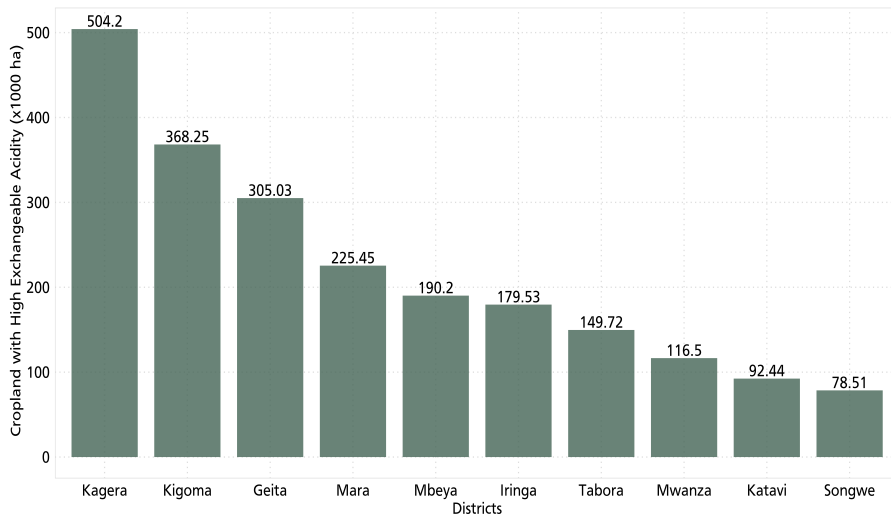
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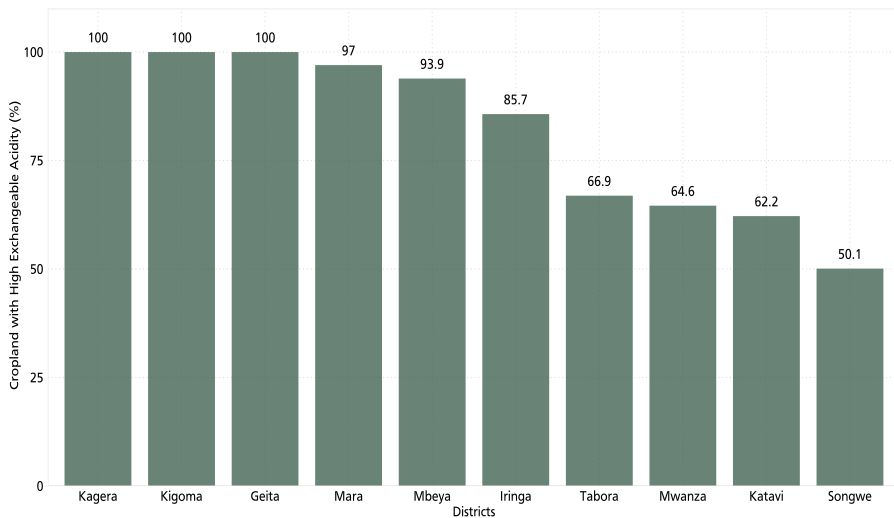
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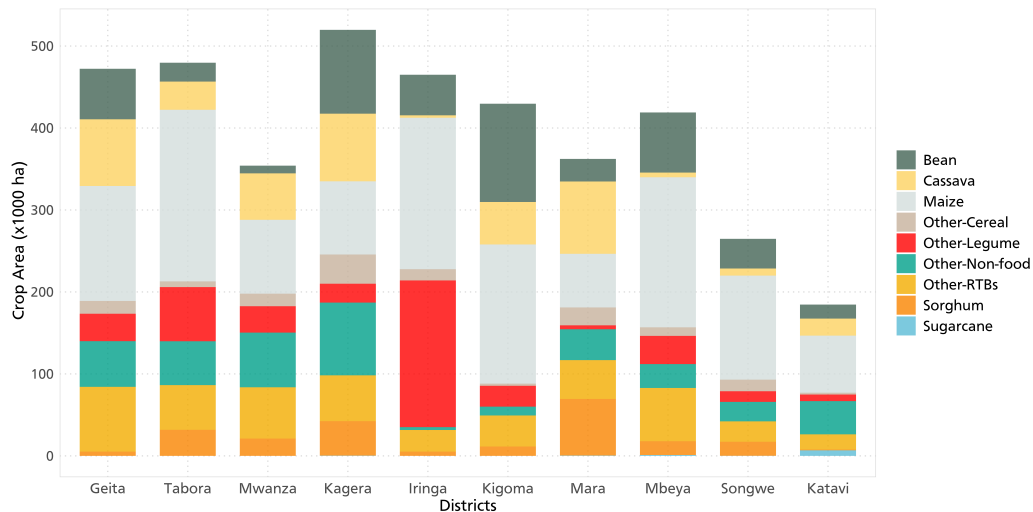
Top 10 Districts in Tanzania with the highest acidic cropland



Top 10 Districts in Tanzania with the highest acidic cropland (Percentage of total cropland)



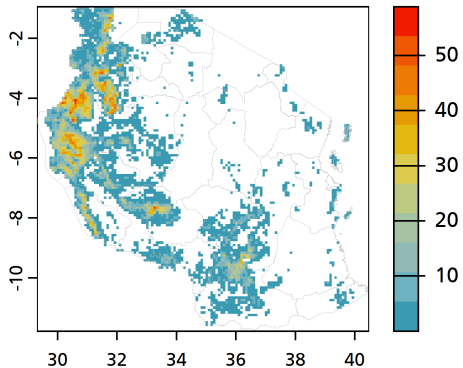
Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland



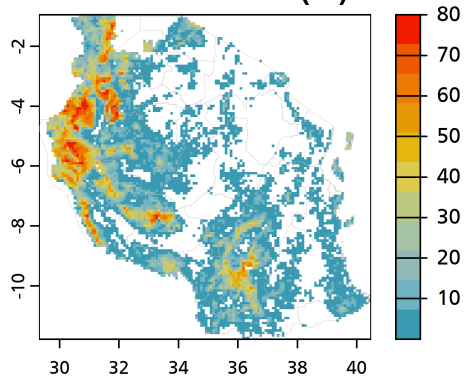
Districts	Maize	Sorghum	Other-Cereal	Bean	Other-Legume	Cassava	Sugarcane	Other-RTBs	Other-Non-food
Geita	140.31	5.21	15.57	61.62	33.46	81.38	-	79.03	55.71
Iringa	184.81	5.18	13.70	49.55	179.01	2.95	0.02	26.34	3.47
Kagera	89.30	42.02	35.70	102.26	22.96	82.45	0.32	55.95	88.77
Katavi	69.70	0.89	2.14	17.08	8.02	20.77	6.81	18.60	40.56
Kigoma	169.76	11.38	2.62	119.90	25.52	51.83	0.04	37.86	10.73
Mara	65.43	69.10	21.82	27.52	4.85	88.24	0.39	47.33	37.64
Mbeya	183.11	16.78	10.51	73.37	34.35	5.62	1.08	65.01	29.11
Mwanza	90.11	21.13	15.23	9.36	32.22	56.74	-	62.60	66.68
Songwe	126.95	17.15	14.15	36.22	13.00	8.65	0.11	24.90	23.68
Tabora	209.44	31.78	6.93	23.04	66.03	34.40	0.00	54.51	53.56

Yield loss due to soil acidity for Maize and Beans in Tanzania

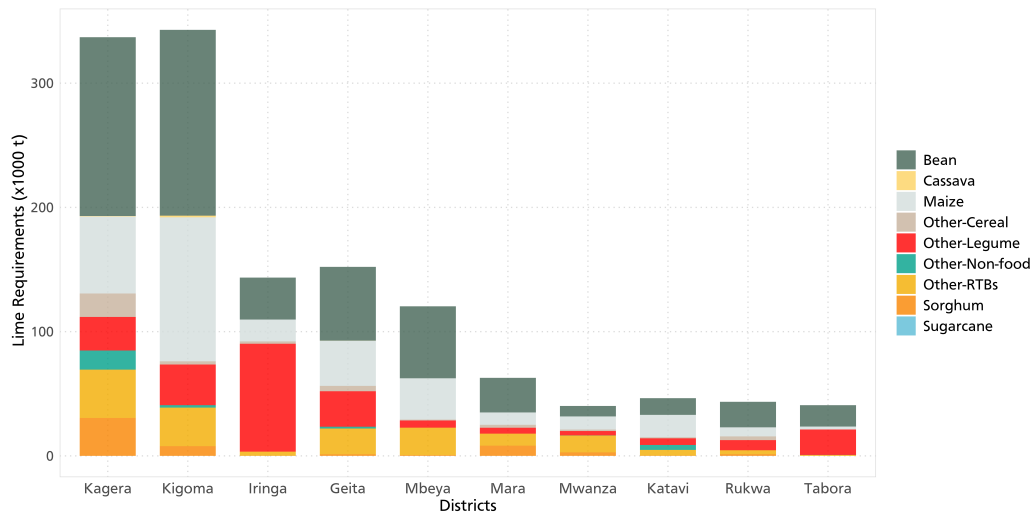
Maize Yield Loss (%)



Bean Yield Loss (%)



Lime requirements (t) for acid soil remediation in Tanzania

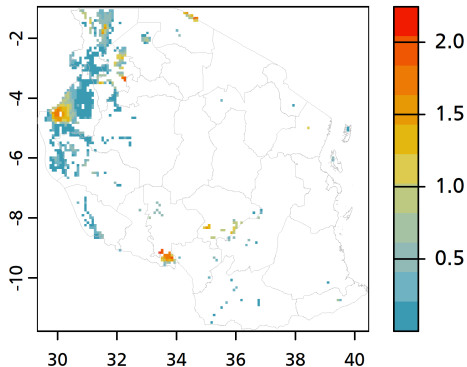


Lime requirements (t) for acid soil remediation in Tanzania (table)

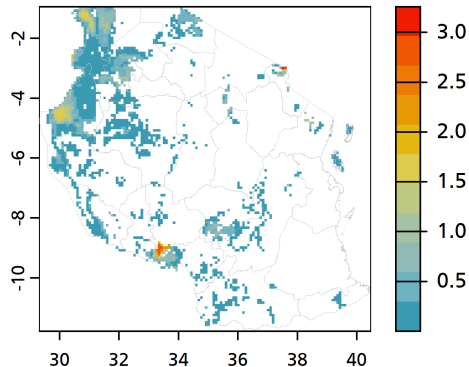
Districts	Maize	Sorghum	Other-Cereal	Bean	Other-Legume	Cassava	Sugarcane	Other-RTBs	Other-Non-food
Geita	36.32	1.27	4.35	59.42	28.57	0.06	-	20.71	1.48
Iringa	17.61	0.16	1.93	33.75	86.78	-	-	3.29	0.00
Kagera	62.01	30.48	18.90	143.91	27.00	0.37	-	38.95	15.39
Katavi	18.01	0.23	0.83	13.40	5.39	0.02	0.01	4.58	3.96
Kigoma	115.89	7.79	2.58	149.50	32.72	1.38	0.00	31.09	1.95
Mara	9.93	8.25	2.42	27.83	4.69	-	-	9.68	-
Mbeya	33.32	0.58	0.68	57.90	5.79	-	-	22.11	0.01
Mwanza	10.55	2.79	1.17	8.48	3.53	-	-	13.60	0.10
Rukwa	7.28	1.36	3.01	20.58	8.17	-	-	3.14	0.00
Tabora	2.34	0.05	0.05	17.22	20.46	0.00	-	0.65	0.06

Lime rate for Maize and Beans in Tanzania

Maize Lime Rate (t/ha)

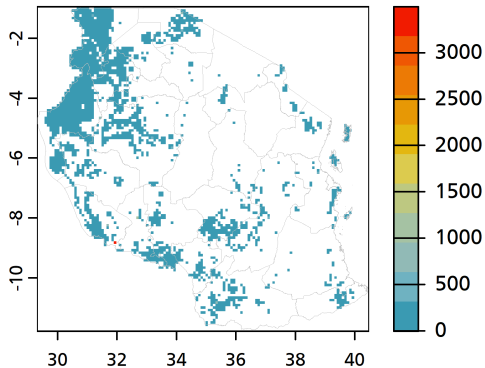


Bean Lime Rate (t/ha)

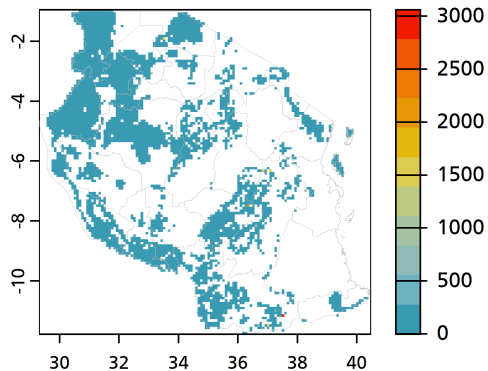


Yield response to lime application in Tanzania

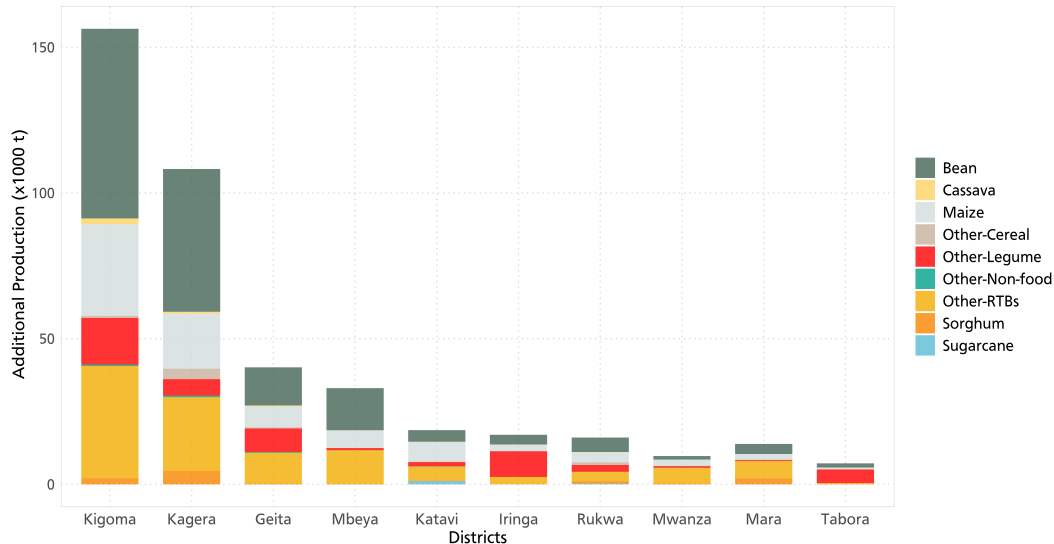
Maize Yield Response (t/ha)



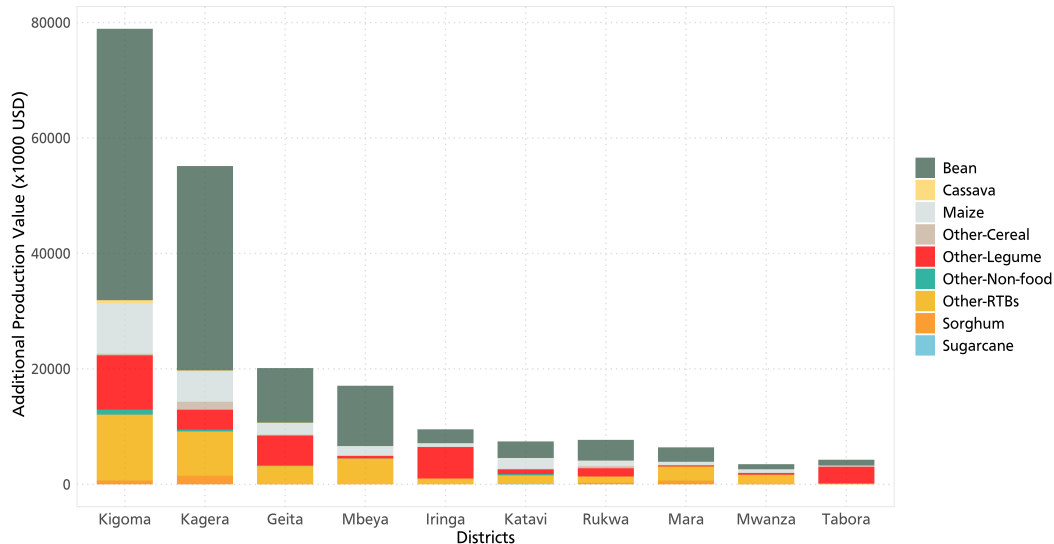
Bean Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Districts in Tanzania

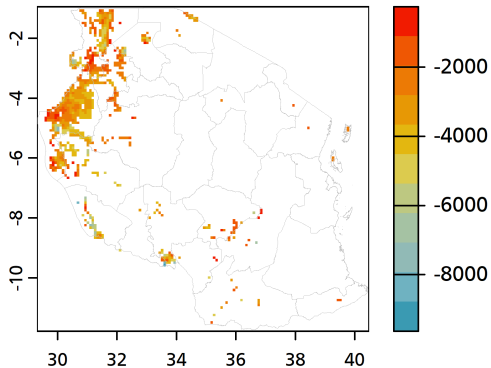


Potential for additional production value (US\$) in top 10 acidic Districts in Tanzania

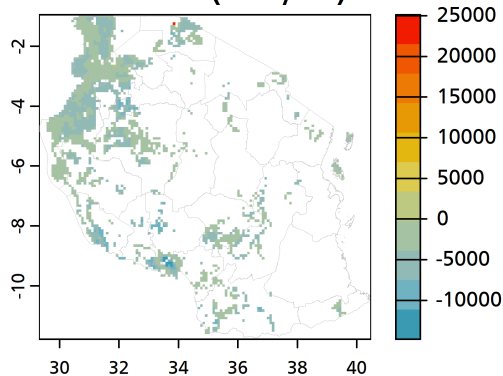


Return on investment for lime application in Tanzania

Maize ROI (USD/ha)

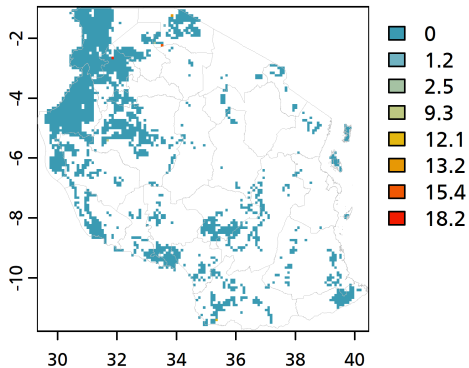


Bean ROI (USD/ha)

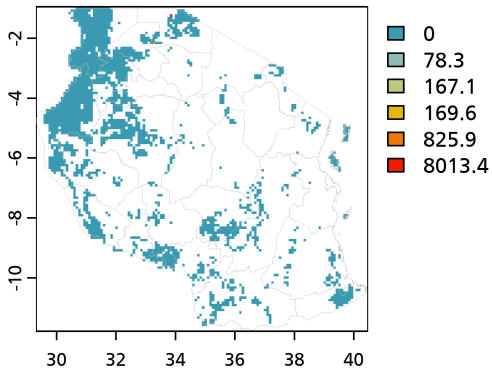


Profitability of lime application in Tanzania

Profitable area (%)



Weighted profit (USD/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop prices are spatially constant; the median crop price across all countries for the period 2016–2020 was used throughout the analysis.
- Lime prices are assumed to be \$100 per tonne.
- Profitability is calculated only for the first year.